

NORTH SYDNEY GIRLS HIGH SCHOOL



2017 TRIAL HSC EXAMINATION

Mathematics Extension 2

General Instructions

- Reading Time – 5 minutes
- Working Time – 3 hours
- Write using black pen
- Board approved calculators may be used
- A reference sheet has been provided
- In Questions 11 – 16, show relevant mathematical reasoning and/or calculations

Total marks – 100

Section I

Pages 2 – 6

10 marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II

Pages 7 – 15

90 Marks

- Attempt Questions 11 – 16
- Allow about 2 hours and 45 minutes for this section

NAME: _____

TEACHER: _____

STUDENT NUMBER: _____

Question	1-10	11	12	13	14	15	16	Total
Mark	/10	/15	/15	/15	/15	/15	/15	/100

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

1 If k is a real number, then what is $\frac{k^2 + 4}{2 - ki}$ equal to?

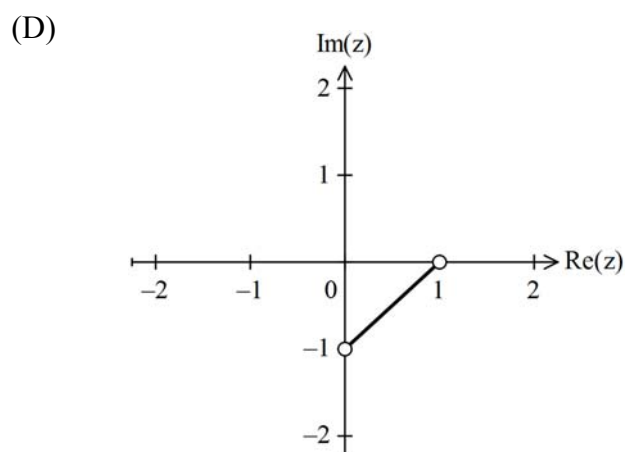
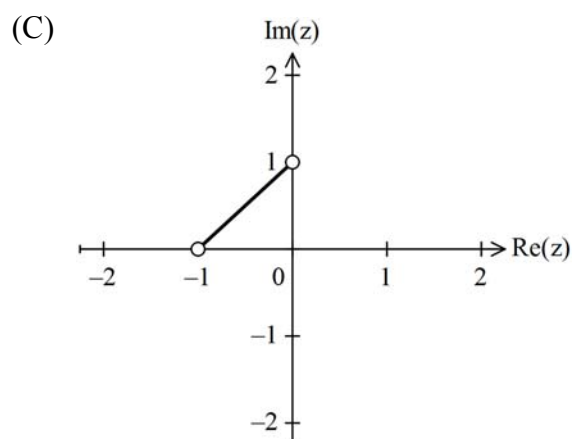
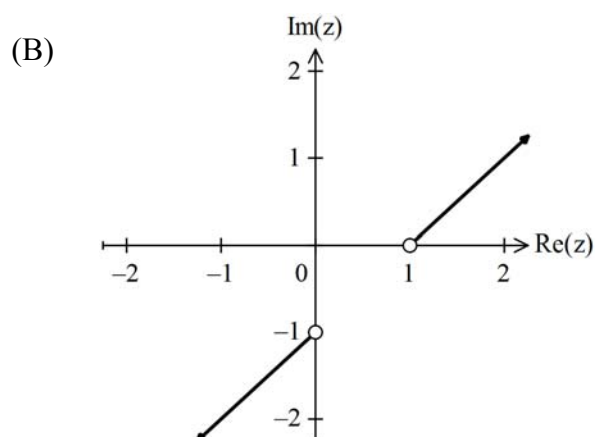
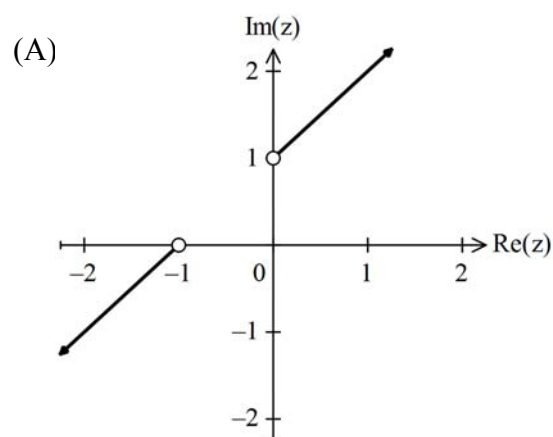
(A) $2 - ki$

(B) $k - 2i$

(C) $2 + ki$

(D) $k + 2i$

2 Which of the following Argand diagrams describes the locus defined by $\arg(z - i) = \arg(z + 1)$?



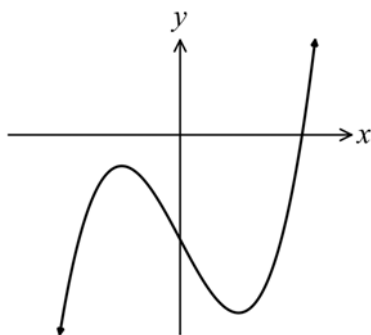
- 3 The roots of the polynomial $x^3 - 2x^2 + 5x + 4 = 0$ are α, β and γ .

What is the value of $\alpha^2 + \beta^2 + \gamma^2$?

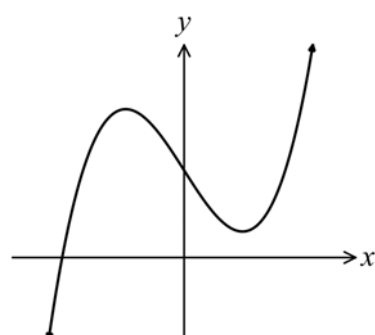
- (A) 4
- (B) 25
- (C) -6
- (D) 14

- 4 The polynomial $P(x) = x^3 - 11x - 20$ has a zero at $x = -2 + i$. Which of the graphs below could be the graph of $y = P(x)$?

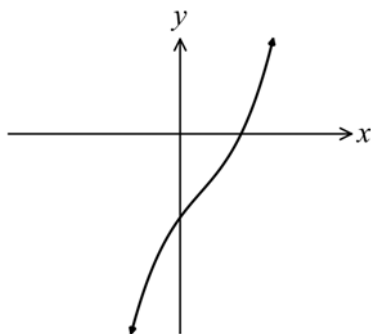
(A)



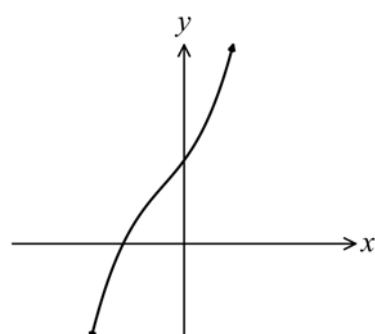
(B)



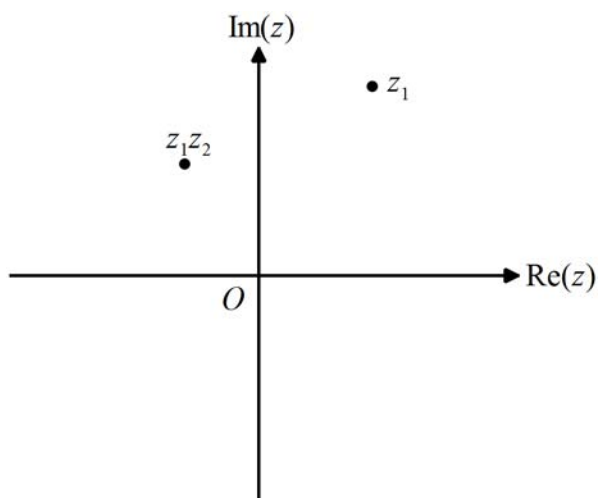
(C)



(D)



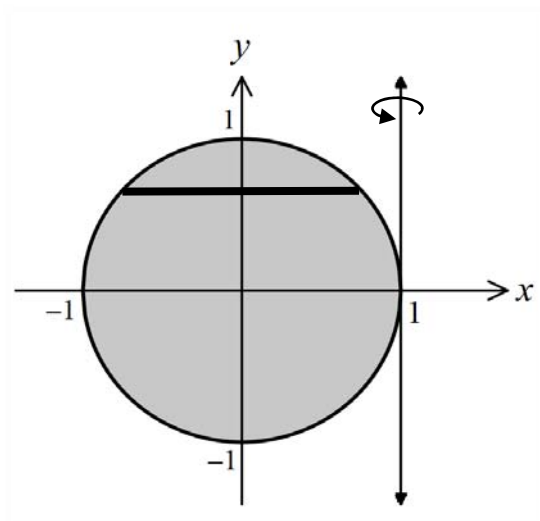
- 5 Let $z_1 = r_1 \text{cis} \alpha$ and $z_2 = r_2 \text{cis} \beta$ where r_1 and r_2 are real numbers. z_1 and $z_1 z_2$ are shown in the Argand diagram below.



Which of the following is necessarily true?

- (A) $r_1 > 1$
- (B) $r_2 > 1$
- (C) $\left| \frac{z_1}{z_2} \right| > r_1$
- (D) $r_2 < |z_1 z_2|$
- 6 P is an extremity of the minor axis of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ with foci at S and S' . If SPS' is a right-angled triangle, what is the eccentricity of the ellipse?
- (A) $\frac{1}{2}$
- (B) $\frac{1}{\sqrt{2}}$
- (C) $\frac{\sqrt{3}}{2}$
- (D) $\frac{2}{\sqrt{3}}$

- 7 The region bounded by the circle $x^2 + y^2 = 1$ is rotated about the line $x = 1$.



What is the volume of the solid of revolution formed?

- (A) $V = 4\pi \int_0^1 (1 - y^2)^{\frac{1}{2}} dy$
- (B) $V = 8\pi \int_0^1 (1 - y^2)^{\frac{1}{2}} dy$
- (C) $V = 4\pi \int_0^1 (1 - y^2) dy$
- (D) $V = 8\pi \int_0^1 (1 - y^2) dy$

- 8 What is the range of the function $f(x) = (x^2 - 1)\sin^{-1}(x - 1)$?

- (A) $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$
- (B) $\frac{\pi}{2} \leq y \leq \frac{3\pi}{2}$
- (C) $0 \leq y \leq \frac{\pi}{2}$
- (D) $0 \leq y \leq \frac{3\pi}{2}$

9 If $a > b$ and $k < 0$, which of the following must be true?

I. $a^2 > b^2$

II. $a + k > b + k$

III. $\frac{a}{k^2} > \frac{b}{k^2}$

(A) I and II only

(B) II and III only

(C) I and III only

(D) I, II and III

10 Which expression must be equal to $\int_0^a [f(a-x) + f(a+x)] dx$?

(A) $\int_0^a f(x) dx$

(B) $\int_0^{2a} f(x) dx$

(C) $2 \int_0^a f(x) dx$

(D) $\int_{-a}^a f(x) dx$

Section II

Total marks – 90

Attempt Questions 11–16

Allow about 2 hour 45 minutes for this section.

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

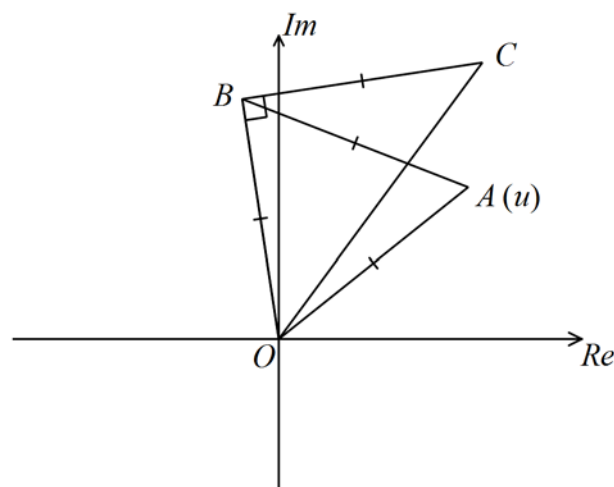
In Questions 11 to 16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet.

- (a) Consider the complex number $z = \sqrt{3} - 3i$.
- (i) Express z in modulus-argument form. 2
 - (ii) Write down the argument of z^4 . 1
 - (iii) Hence, write z^4 in the form $x + iy$, where x and y are real. 1
- (b) Find $\int \sin^3 x \, dx$. 2
- (c) Sketch the conic defined by $|z - 1| + |z + 1| = 4$ showing intercepts, foci and directrices. 2
- (d) Consider the polynomial equation $P(z) = z^4 - 4z^3 + 7z^2 - 4z + 6$.
- (i) Show that $z = i$ is a zero of this polynomial. 1
 - (ii) Hence, write down a quadratic real factor of $P(z)$. 1
 - (iii) Find all the roots of $P(z) = 0$. 2
- (e) A relation is defined by the equation $\sin x + \cos y = \frac{1}{2}$,
where $-\pi < x < \pi$ and $-\pi < y < \pi$.
- (i) Show that $\frac{dy}{dx} = \frac{\cos x}{\sin y}$. 1
 - (ii) Find the coordinates of the points where $\frac{dy}{dx} = 0$. 2

Question 12 (15 marks) Use a SEPARATE writing booklet.

(a)



In the Argand diagram above, $OA = OB = AB = BC$ and $OB \perp BC$.
 A represents the complex number u .

- (i) Find the complex number represented by B . 1
- (ii) Hence, or otherwise, find an expression in terms of u for the complex number represented by C . 2

- (b) (i) Find real numbers a , b and c such that 2

$$\frac{3-x}{(x^2+1)(1-2x)} = \frac{ax+b}{x^2+1} + \frac{c}{1-2x}.$$

- (ii) Hence find $\int \frac{3-x}{(x^2+1)(1-2x)} dx$. 2

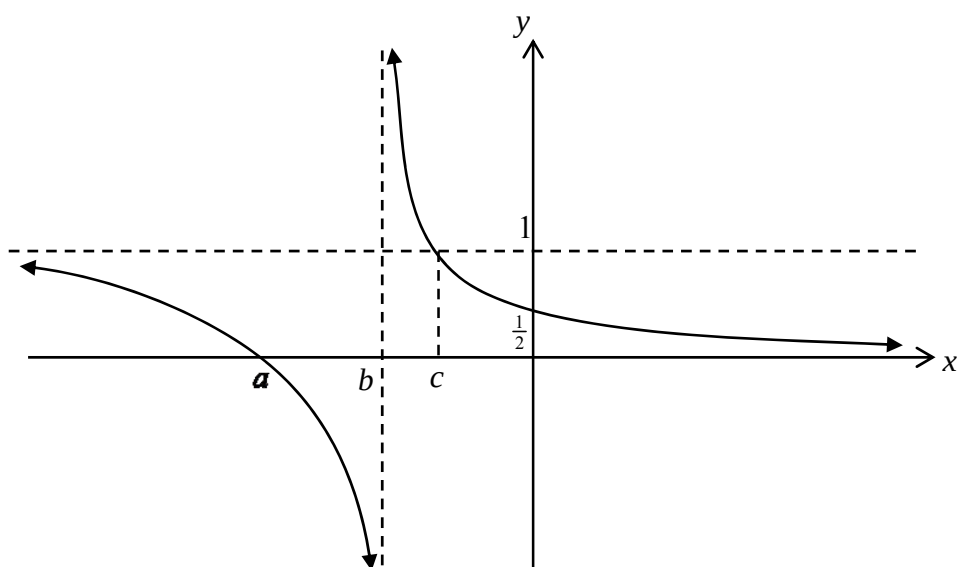
- (c) (i) Sketch the curve $f(x) = \frac{x-3}{x^2+x-2}$ showing all intercepts and asymptotes. 3
 You are not required to find stationary points.

- (ii) Hence solve $\frac{|x|-3}{x^2+|x|-2} \geq 0$. 1

Question 12 continues on Page 9

Question 12 (continued)

- (d) Given below is the graph of $y = f(x)$. The line $y = 1$ is a horizontal asymptote and $x = b$ is a vertical asymptote. The x -intercept is at $x = a$ and the y -intercept is at $y = \frac{1}{2}$.



Neatly sketch the graphs of the following showing all important information, including the location of a, b and c :

- | | | |
|------|-------------------|---|
| (i) | $\frac{1}{f(x)}$ | 2 |
| (ii) | $\tan^{-1}(f(x))$ | 2 |

End of Question 12

Question 13 (15 marks) Use a SEPARATE writing booklet.

(a) If $u = 6 + ki$ and $v = 4 + ki$, find k if $\arg(uv) = \frac{\pi}{4}$. 3

(b) $P(a \sec \theta, b \tan \theta)$ is a point on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.

(i) Show that the tangent at P has the equation 2

$$\frac{x \sec \theta}{a} - \frac{y \tan \theta}{b} = 1$$

(ii) The tangent at P meets the asymptotes of the hyperbola at Q and R . 3
Show that $PQ = PR$.

(c) If α, β, γ are the roots of the equation $x^3 + 2x^2 + 3x - 4 = 0$, find a polynomial equation whose roots are:

(i) $\frac{1}{\alpha}, \frac{1}{\beta}$ and $\frac{1}{\gamma}$ 2

(ii) α^2, β^2 and γ^2 2

(d) A sequence is defined by 3
 $a_1 = 5$, $a_2 = 13$ and $a_{n+2} = 5a_{n+1} - 6a_n$ for all natural numbers n .

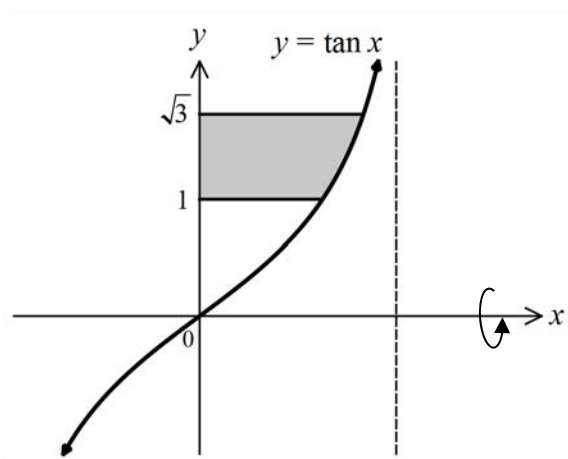
Use Mathematical Induction to prove that $a_n = 2^n + 3^n$.

Question 14 (15 marks) Use a SEPARATE writing booklet

- (a) Using the substitution $t = \tan \frac{x}{2}$, or otherwise, evaluate 3

$$\int_0^{\frac{\pi}{3}} \frac{1}{1 + \cos x - \sin x} dx .$$

- (b) (i) Use integration by parts to find $\int x \tan^{-1} x \, dx$. 2
- (ii) The region bounded by the curve $y = \tan x$, the lines $y = 1$, $y = \sqrt{3}$ and the y -axis is rotated about the x -axis. 2



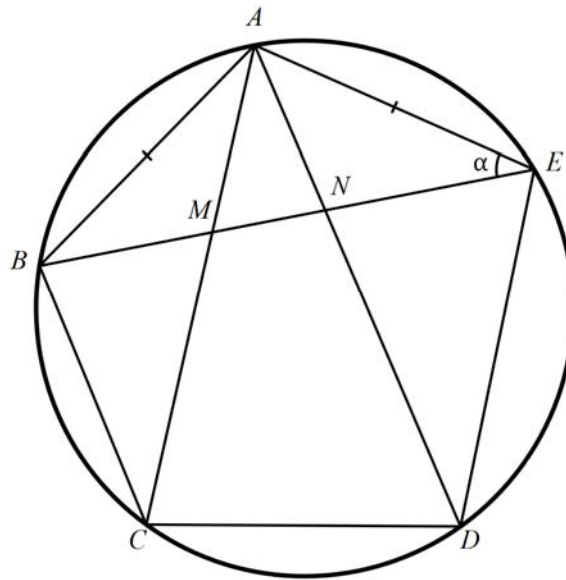
Use the method of cylindrical shells to find the volume of the solid of revolution formed.

- (c) (i) Show that $a^2 + 9b^2 \geq 6ab$, where a and b are real numbers. 1
- (ii) Hence, or otherwise, show that $a^2 + 5b^2 + 9c^2 \geq 3(ab + bc + ac)$. 2
- (iii) Hence if $a > b > c > 0$, show that $a^2 + 5b^2 + 9c^2 > 9bc$. 1

Question 14 continues on Page 12

Question 14 (continued)

- (d) $ABCDE$ is a pentagon inscribed in a circle and $AB = AE$.
 BE meets AC and AD at M and N respectively. Let $\angle BEA = \alpha$.



- | | | |
|------|--|----------|
| (i) | Explain why $\angle ACE = \alpha$. | 1 |
| (ii) | Prove that $CDNM$ is a cyclic quadrilateral. | 3 |

End of Question 14

Question 15 (15 marks) Use a SEPARATE writing booklet.

- (a) Given that $p(x) = x^3 - 3ax^2 + b$ has a double zero, where a and b are non-zero real numbers, show that $4a^3 - b = 0$. 2

- (b) $P\left(cp, \frac{c}{p}\right)$ lies on the hyperbola $xy = c^2$.

- (i) Show that the equation of the normal to the hyperbola at P is given by 2

$$px - \frac{y}{p} = c\left(p^2 - \frac{1}{p^2}\right).$$

- (ii) The equation of the tangent at P is $x + p^2y = 2cp$. DO NOT PROVE THIS. 3

The normal at P cuts the x -axis at Q and the tangent at P cuts the y -axis at R .
 M is the midpoint of QR .

Find the equation of the locus of M as P moves on the hyperbola.

- (c) Let $I = \int_{\frac{1}{a}}^a \frac{f(x)}{x\left(f(x) + f\left(\frac{1}{x}\right)\right)} dx$.

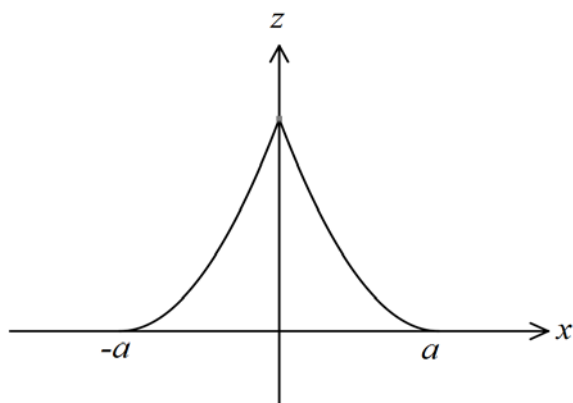
- (i) Use a suitable substitution to show that $I = \int_{\frac{1}{a}}^a \frac{f\left(\frac{1}{x}\right)}{x\left(f(x) + f\left(\frac{1}{x}\right)\right)} dx$. 2

- (ii) Hence, or otherwise, evaluate $\int_{\frac{1}{2}}^2 \frac{\sin x}{x\left(\sin x + \sin \frac{1}{x}\right)} dx$. 2

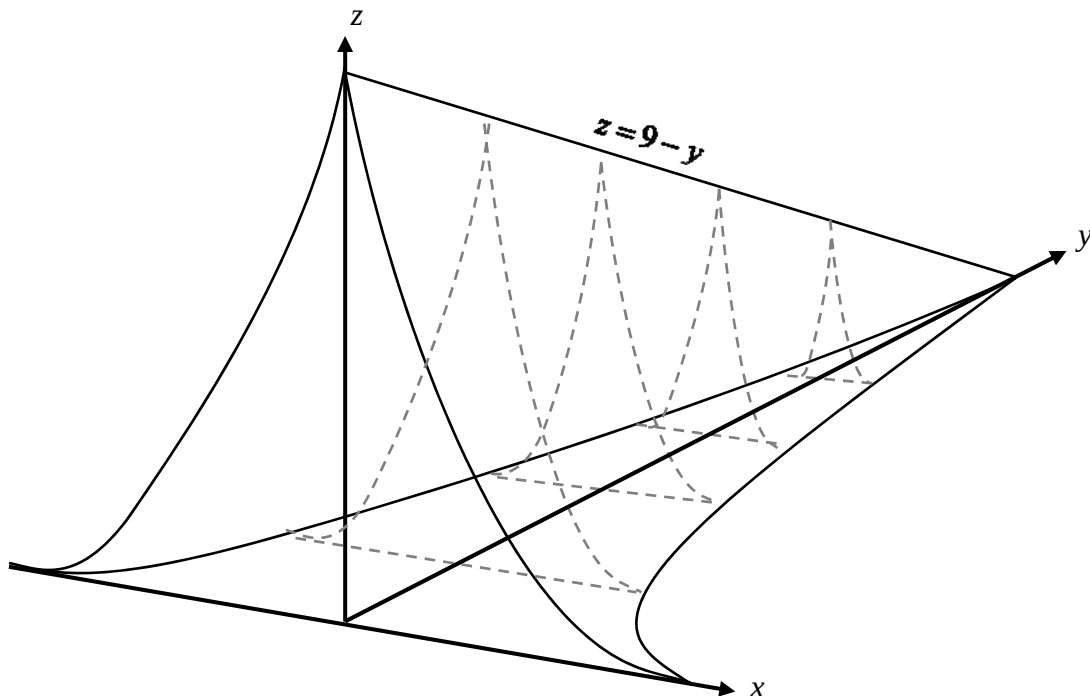
Question 15 continues on Page 14

Question 15 (continued)

- (d) (i) Find the area bounded by the curve $y = (x - a)^2$ and the coordinate axes, where $a > 0$. 1
- (ii) Cross-sections of a solid perpendicular to the base are sections of two parabolas $z = (x - a)^2$ and $z = (x + a)^2$ as shown below. 3



The heights of the cross-sections are bounded by the line $z = 9 - y$.



Find the volume of the solid.

End of Question 15

Question 16 (15 marks) Use a SEPARATE writing booklet.

(a) Consider $I_n = \int (x^2 + 1)^{-n} dx$, $n > 0$.

(i) Show that $I_{n+1} = \frac{x(x^2 + 1)^{-n}}{2n} + \frac{2n-1}{2n} I_n$. 3

(ii) Find I_2 . 1

(b) (i) Using the result $\cos(A+B) - \cos(A-B) = -2\sin A \sin B$, or otherwise, 2

show that $\sin \theta \sum_{r=1}^n \sin 2r\theta = \sin(n+1)\theta \sin n\theta$.

(ii) Hence show that $\sum_{r=1}^8 \sin \frac{r\pi}{9} = \cot \frac{\pi}{18}$. 2

(c) (i) Show that $\log_e x \leq x - 1$ for $x > 0$. 2

(ii) Show that $\log_e \left(\frac{c_1 c_2 \dots c_n}{\mu^n} \right) \leq \frac{c_1 + c_2 + \dots + c_n}{\mu} - n$, where $c_1, c_2, \dots, c_n > 0$ and $\mu > 0$. 2

(iii) Hence if $\mu = \frac{c_1 + c_2 + \dots + c_n}{n}$, show that $\sqrt[n]{c_1 c_2 \dots c_n} \leq \frac{c_1 + c_2 + \dots + c_n}{n}$. 2

(iv) Hence use part (iii) to find a lower bound for $\frac{101}{103} + \frac{103}{105} + \frac{105}{107} + \dots + \frac{197}{199} + \frac{199}{101}$. 1

End of paper

- 1 If k is a real number, then what is $\frac{k^2 + 4}{2 - ki}$ equal to?

(A) $2 - ki$

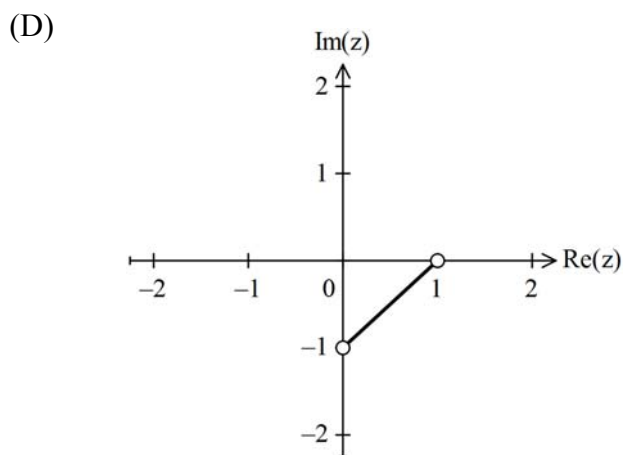
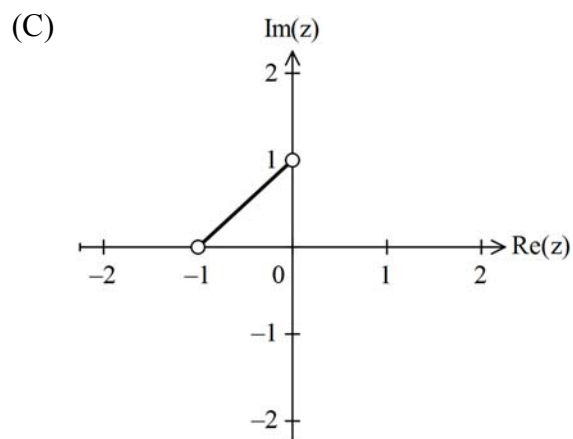
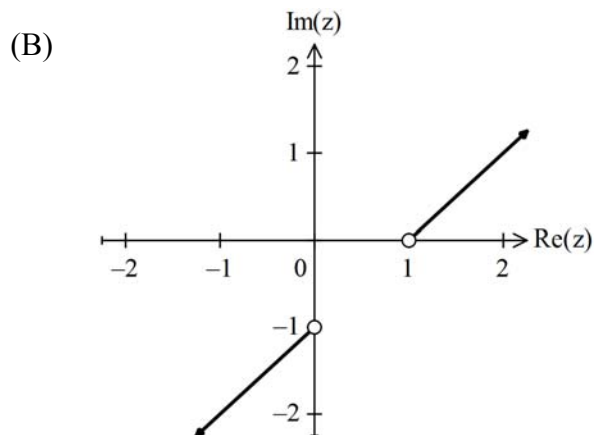
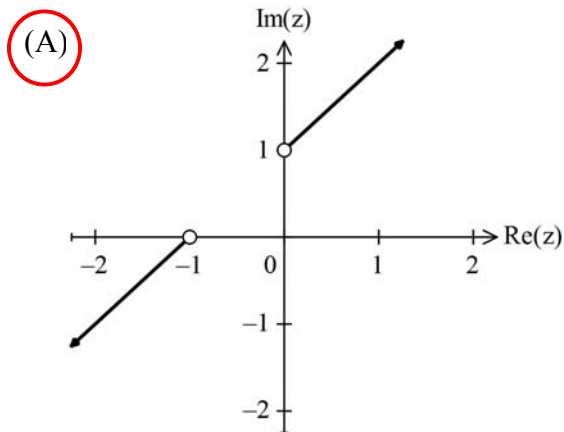
(B) $k - 2i$

(C) $2 + ki$

(D) $k + 2i$

$$\frac{k^2 + 4}{2 - ki} \times \frac{2 + ki}{2 + ki} = \frac{\cancel{k^2 + 4} \times (2 + ki)}{\cancel{4 + k^2}} = 2 + ki$$

- 2 Which of the following Argand diagrams describes the locus defined by $\arg(z - i) = \arg(z + 1)$?



- 3 The roots of the polynomial $x^3 - 2x^2 + 5x + 4 = 0$ are α, β and γ .

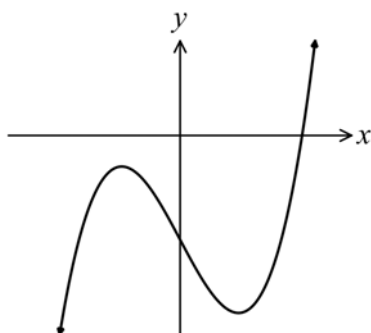
What is the value of $\alpha^2 + \beta^2 + \gamma^2$?

- (A) 4
(B) 25
(C) -6
(D) 14

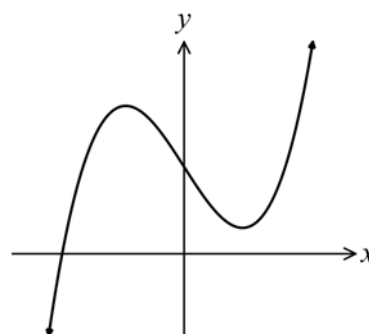
$$\begin{aligned}\alpha^2 + \beta^2 + \gamma^2 &= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha) \\ &= (-2)^2 - 2(5) = 4 - 10 \\ &= -6\end{aligned}$$

- 4 The polynomial $P(x) = x^3 - 11x - 20$ has a zero at $x = -2 + i$. Which of the graphs below could be the graph of $y = P(x)$?

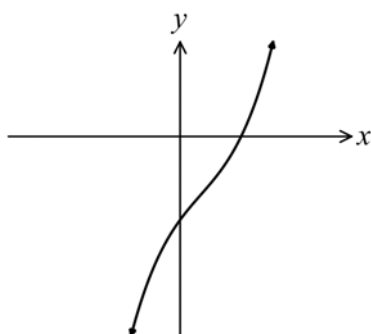
(A)



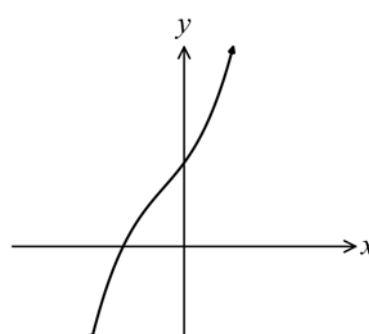
(B)



(C)



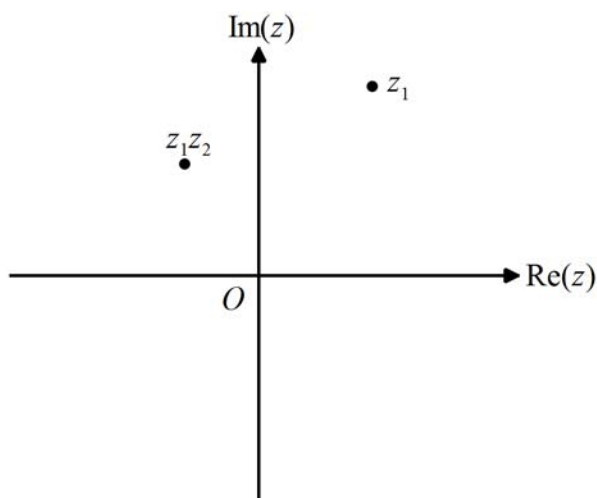
(D)



The graph has a negative y-intercept so must be A or C. Solving for stationary points,

$$P'(x) = 0 \Rightarrow 3x^2 - 11 = 0 \Rightarrow x = \pm\sqrt{\frac{11}{3}}. \text{ So it has stationary points. Hence, A.}$$

- 5 Let $z_1 = r_1 \text{cis} \alpha$ and $z_2 = r_2 \text{cis} \beta$ where r_1 and r_2 are real numbers. z_1 and $z_1 z_2$ are shown in the Argand diagram below.



Which of the following is necessarily true?

(A) $r_1 > 1$

(B) $r_2 > 1$

(C) $\left| \frac{z_1}{z_2} \right| > r_1$

(D) $r_2 < |z_1 z_2|$

There is no scale, so neither r_1 or r_2 needs to be greater than 1. D looks like it has to be true, but if both r_1 and r_2 are smaller than 1, then $|z_1 z_2| = r_1 r_2$ will be smaller than r_2 . Rearranging C we get $|z_1| > r_1 |z_2|$ or $|z_1| > |z_1 z_2|$ which is true from the diagram. Hence, C.

- 6 P is an extremity of the minor axis of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ with foci at S and S' . If SPS' is a right-angled triangle, what is the eccentricity of the ellipse?

(A) $\frac{1}{2}$

(B) $\frac{1}{\sqrt{2}}$

(C) $\frac{\sqrt{3}}{2}$

(D) $\frac{2}{\sqrt{3}}$

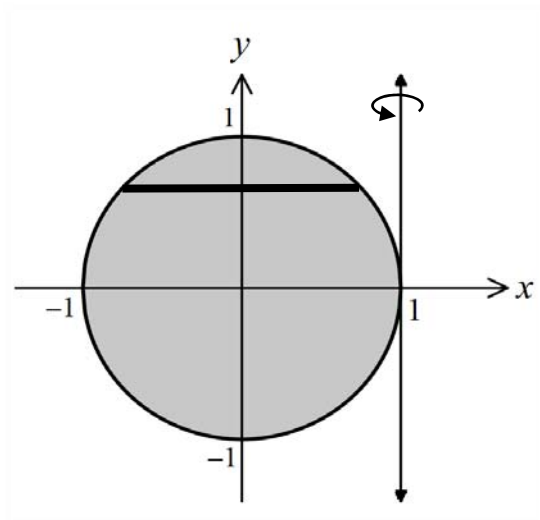
$P = (0, \pm b)$ and $m_{PS} \times m_{PS'} = -1$

$$\frac{b-0}{0-ae} \times \frac{b-0}{0+ae} = -1 \Rightarrow \frac{b^2}{-a^2 e^2} = -1$$

$$b^2 = a^2 e^2 \Rightarrow e^2 = \frac{b^2}{a^2}$$

$$e^2 = 1 - e^2 \text{ or } 2e^2 = 1 \Rightarrow e = \frac{1}{\sqrt{2}}$$

- 7 The region bounded by the circle $x^2 + y^2 = 1$ is rotated about the line $x = 1$.



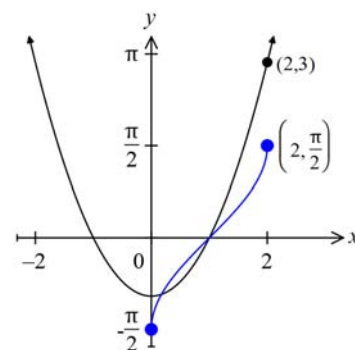
What is the volume of the solid of revolution formed?

- (A) $V = 4\pi \int_0^1 (1 - y^2)^{\frac{1}{2}} dy$
- (B) $V = 8\pi \int_0^1 (1 - y^2)^{\frac{1}{2}} dy$
- (C) $V = 4\pi \int_0^1 (1 - y^2) dy$
- (D) $V = 8\pi \int_0^1 (1 - y^2) dy$

$$\begin{aligned}
 x^2 + y^2 = 1 &\Rightarrow x = \pm\sqrt{1 - y^2} \\
 R &= 1 + \sqrt{1 - y^2} \text{ and } r = 1 - \sqrt{1 - y^2} \\
 V &= \pi \int_{-1}^1 (R^2 - r^2) dy \\
 &= 2\pi \int_0^1 (R + r)(R - r) dy \\
 &= 2\pi \int_0^1 (2)(2\sqrt{1 - y^2}) dy \\
 &= 8\pi \int_0^1 (1 - y^2)^{\frac{1}{2}} dy
 \end{aligned}$$

- 8 What is the range of the function $f(x) = (x^2 - 1)\sin^{-1}(x - 1)$?

- (A) $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$
- (B) $\frac{\pi}{2} \leq y \leq \frac{3\pi}{2}$
- (C) $0 \leq y \leq \frac{\pi}{2}$
- (D) $0 \leq y \leq \frac{3\pi}{2}$



From the graphs of $y = x^2 - 1$ and $y = \sin^{-1}(x - 1)$ both are negative or both are positive. So the range is always positive. When both are 0, $y = 0$, so not B.

Finally, looking at endpoints of the domain of $y = \sin^{-1}(x - 1)$, the answer is D.

- 9 If $a > b$ and $k < 0$, which of the following must be true?

- I. $a^2 > b^2$
II. $a + k > b + k$
III. $\frac{a}{k^2} > \frac{b}{k^2}$

- (A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

I is true only if a and b are positive.

II is true regardless of whether a, b , or k are positive.

III is true as $k^2 > 0$ when $k < 0$

Therefore II and III are true, so B.

- 10 Which expression must be equal to $\int_0^a [f(a-x) + f(a+x)] dx$?

- (A) $\int_0^a f(x) dx$
(B) $\int_0^{2a} f(x) dx$
(C) $2 \int_0^a f(x) dx$
(D) $\int_{-a}^a f(x) dx$

$$\int_0^a f(a-x) dx = \int_0^a f(x) dx \text{ by symmetry in } x = \frac{a}{2}.$$

$y = f(a+x)$ is $y = f(x)$ shifted right by a units.

$$\int_0^a f(a+x) dx = \int_a^{2a} f(x) du.$$

Hence,

$$\begin{aligned} \int_0^a [f(a-x) + f(a+x)] dx &= \int_0^a f(x) dx + \int_a^{2a} f(x) dx \\ &= \int_0^{2a} f(x) dx \end{aligned}$$

Question 11 (15 marks) Use a SEPARATE writing booklet.

(a) Consider the complex number $z = \sqrt{3} - 3i$.

- | | | |
|-------|--|---|
| (i) | Express z in modulus-argument form. | 2 |
| (ii) | Write down the argument of z^4 . | 1 |
| (iii) | Hence write z^4 in the form $x + iy$, where x and y are real. | 1 |

$$(i) \quad |z| = \sqrt{\sqrt{3}^2 + (-3)^2} = \sqrt{12}$$

$$\tan(\arg z) = \frac{-3}{\sqrt{3}} = -\sqrt{3} \Rightarrow \arg z = -\frac{\pi}{3} \text{ (4th quadrant)}$$

$$z = 2\sqrt{3} \left(\cos\left(-\frac{\pi}{3}\right) + i \sin\left(-\frac{\pi}{3}\right) \right)$$

$$(ii) \quad \arg(z^4) = 4 \arg(z) = 4 \left(-\frac{\pi}{3} \right) = -\frac{4\pi}{3} \text{ or } \frac{2\pi}{3}$$

$$\begin{aligned} (iii) \quad z^4 &= (2\sqrt{3})^4 \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right) \\ &= 144 \left(-\frac{1}{2} + i \frac{\sqrt{3}}{2} \right) \\ &= -72 + 72\sqrt{3}i \end{aligned}$$

(b) Find $\int \sin^3 x \, dx$.

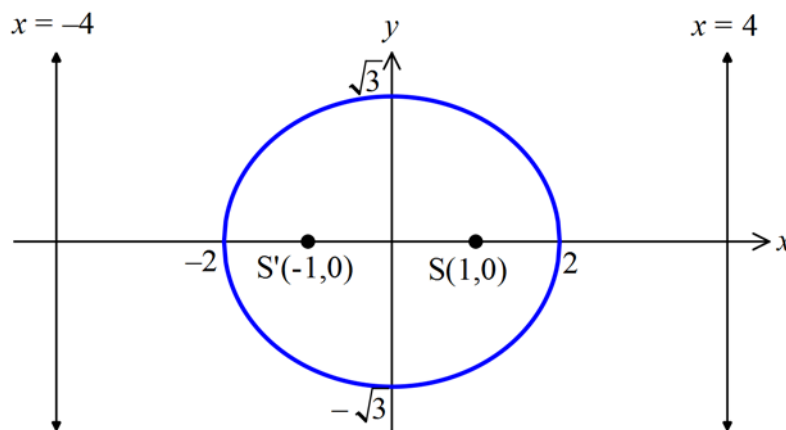
2

$$\begin{aligned} \int \sin^3 x \, dx &= \int (1 - \cos^2 x) \sin x \, dx \\ &= \int \sin x \, dx + \int \cos^2 x (-\sin x) \, dx \\ &= -\cos x + \frac{\cos^3 x}{3} + C \end{aligned}$$

(c) Sketch the conic defined by $|z-1|+|z+1|=4$ showing intercepts, foci and directrices. 2

Using $PS + PS' = 2a$, this is an ellipse with $2a = 4$ and $S(1,0)$ and $S'(-1,0)$.

Then $a = 2$; and $ae = 1 \Rightarrow e = \frac{1}{2}$. Then $\frac{a}{e} = 4$ and $b^2 = a^2(1 - e^2) = 4\left(1 - \frac{1}{4}\right) = 3$



(d) Consider the polynomial equation $P(z) = z^4 - 4z^3 + 7z^2 - 4z + 6$.

(i) Show that $z = i$ is a zero of this polynomial. 1

(ii) Hence write down a quadratic real factor of $P(z)$. 1

(iii) Find all the roots of $P(z) = 0$. 2

(i) $P(i) = i^4 - 4i^3 + 7i^2 - 4i + 6 = 1 + 4i - 7 - 4i + 6 = 0$. Therefore, $z = i$ is a zero or $z = i$ is a root of $P(z) = 0$.

(ii) As the polynomial equation $P(z) = 0$ has real coefficients, complex roots occur in conjugate pairs. Thus, $z = -i$ is also a root and $(z - i)(z + i) = z^2 - i^2 = z^2 + 1$ is a factor of $P(z)$.

(iii) Hence, $P(z) = z^4 - 4z^3 + 7z^2 - 4z + 6 = (z^2 + 1)(z^2 - 4z + 6)$ by inspection

$$z^2 - 4z + 6 = 0$$

$$(z - 2)^2 = -2 \quad \text{by completing the square}$$

$$z - 2 = \pm i\sqrt{2}$$

$$z = 2 \pm i\sqrt{2}$$

Therefore the roots are $\pm i, 2 \pm i\sqrt{2}$.

- (e) A relation is defined by the equation $\sin x + \cos y = \frac{1}{2}$,
where $-\pi < x < \pi$ and $-\pi < y < \pi$.

- (i) Show that $\frac{dy}{dx} = \frac{\cos x}{\sin y}$. 1
- (ii) Find the coordinates of the points where $\frac{dy}{dx} = 0$. 2

- (i) Differentiating implicitly wrt x

$$\cos x - \sin y \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{-\cos x}{-\sin y} = \frac{\cos x}{\sin y}$$

- (ii) $\frac{dy}{dx} = 0 \Rightarrow \cos x = 0$ and $\sin y \neq 0$.

$$\cos x = 0 \Rightarrow x = \pm \frac{\pi}{2} \quad (-\pi < x < \pi).$$

$$x = \frac{\pi}{2} \Rightarrow \sin \frac{\pi}{2} + \cos y = \frac{1}{2}$$

$$x = -\frac{\pi}{2} \Rightarrow \sin \left(-\frac{\pi}{2} \right) + \cos y = \frac{1}{2}$$

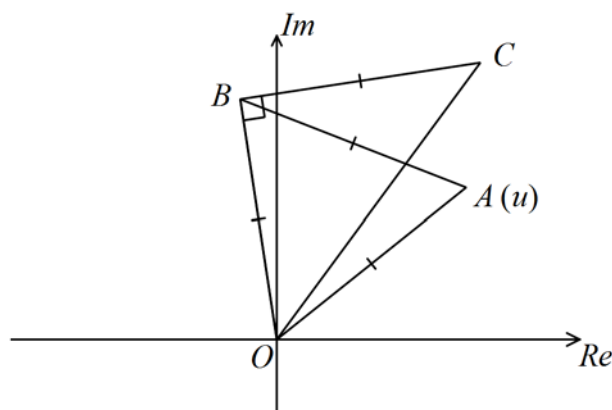
$$\cos y = -\frac{1}{2} \Rightarrow y = \pm \frac{2\pi}{3}$$

$$\cos y = \frac{3}{2} \Rightarrow \text{no solutions}$$

Therefore, $\frac{dy}{dx} = 0$ at $\left(\frac{\pi}{2}, \frac{2\pi}{3} \right)$ and $\left(\frac{\pi}{2}, -\frac{2\pi}{3} \right)$.

Question 12 (15 marks) Use a SEPARATE writing booklet.

(a)



In the Argand diagram above, $OA = OB = AB = BC$ and $OB \perp BC$.
A represents the complex number u .

(i) Find the complex number represented by B. 1

(ii) Hence, or otherwise, find the complex number represented by C. 2

(i) $\angle AOB = \frac{\pi}{3}$ (equilateral triangle)

$\therefore B \equiv u \operatorname{cis}\left(\frac{\pi}{3}\right)$. As $\overline{OB}$ is obtained by rotating $\overline{OA}$ anticlockwise by $\frac{\pi}{3}$.

(ii) $\overline{OC} = \overline{OB} + \overline{BC}$

Now, $\overline{BC}$ is obtained by rotating $\overline{BO}$ anticlockwise by $\frac{\pi}{2}$.

$$\therefore C \equiv u \operatorname{cis}\left(\frac{\pi}{3}\right) - i \cdot u \operatorname{cis}\left(\frac{\pi}{3}\right)$$

$$C \equiv (1 - i)u \operatorname{cis}\left(\frac{\pi}{3}\right)$$

Other valid approaches possible. For instance,

$$|\overline{OC}|^2 = |\overline{OB}|^2 + |\overline{BC}|^2 \text{ Pythag. So, } |\overline{OC}| = |u|\sqrt{2}$$

$$\angle AOC = \angle AOB - \angle BOC = \frac{\pi}{3} - \frac{\pi}{4} = \frac{\pi}{12}$$

$\overline{OC}$ is obtained by rotating $\overline{OA}$ anticlockwise by $\frac{\pi}{12}$. So, $\overline{OC} = u\sqrt{2} \operatorname{cis}\left(\frac{\pi}{12}\right)$.

(b) (i) Find real numbers a , b and c such that

2

$$\frac{3-x}{(x^2+1)(1-2x)} = \frac{ax+b}{x^2+1} + \frac{c}{1-2x}.$$

Comparing numerators on both sides, $3-x \equiv (ax+b)(1-2x) + c(x^2+1)$

$$x = \frac{1}{2}: \quad \frac{5}{2} = \frac{5c}{4} \Rightarrow c = 2$$

$$x = 0: \quad 3 = b + c \Rightarrow b = 1$$

$$x^2 \text{coeff: } 0 = -2a + c \Rightarrow a = 1$$

$$\therefore a = 1, \quad b = 1, \quad c = 2$$

(ii) Hence find $\int \frac{3-x}{(x^2+1)(1-2x)} dx$.

2

$$\begin{aligned} \int \frac{3-x}{(x^2+1)(1-2x)} dx &= \int \frac{x+1}{x^2+1} dx + \int \frac{2}{1-2x} dx \quad \text{using (i)} \\ &= \int \frac{x}{x^2+1} dx + \int \frac{1}{x^2+1} dx - \int \frac{-2}{1-2x} dx \\ &= \frac{1}{2} \log_e(x^2+1) + \tan^{-1} x - \log_e(1-2x) + C \end{aligned}$$

(c) (i) Sketch the curve $f(x) = \frac{x-3}{x^2+x-2}$ showing all intercepts and asymptotes.

3

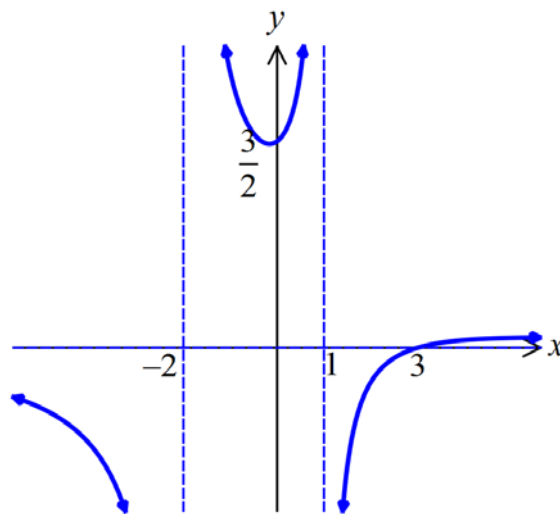
You are not required to find stationary points.

$$f(x) = \frac{x-3}{x^2+x-2} = \frac{x-3}{(x+2)(x-1)}$$

Vertical asymptotes at $x = -2, x = 1$.

$$\lim_{x \rightarrow \pm\infty} f(x) = 0.$$

x -intercept is 3, y -intercept is $\frac{3}{2}$.

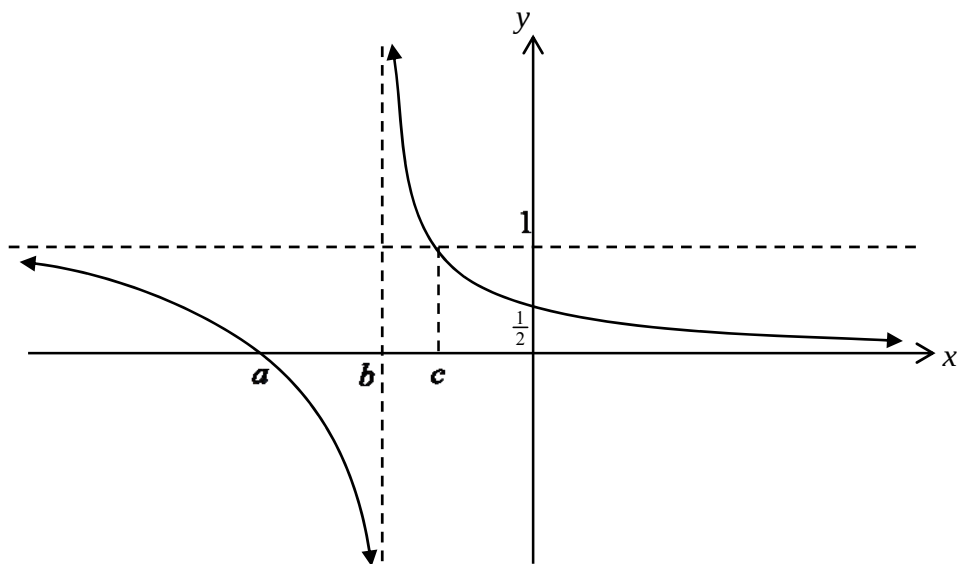


(ii) Hence solve $\frac{|x|-3}{x^2+|x|-2} \geq 0$.

1

From the graph, using the symmetry transformation $y = f(|x|)$, solutions are $x \geq 3, x \leq -3, -1 < x < 1$

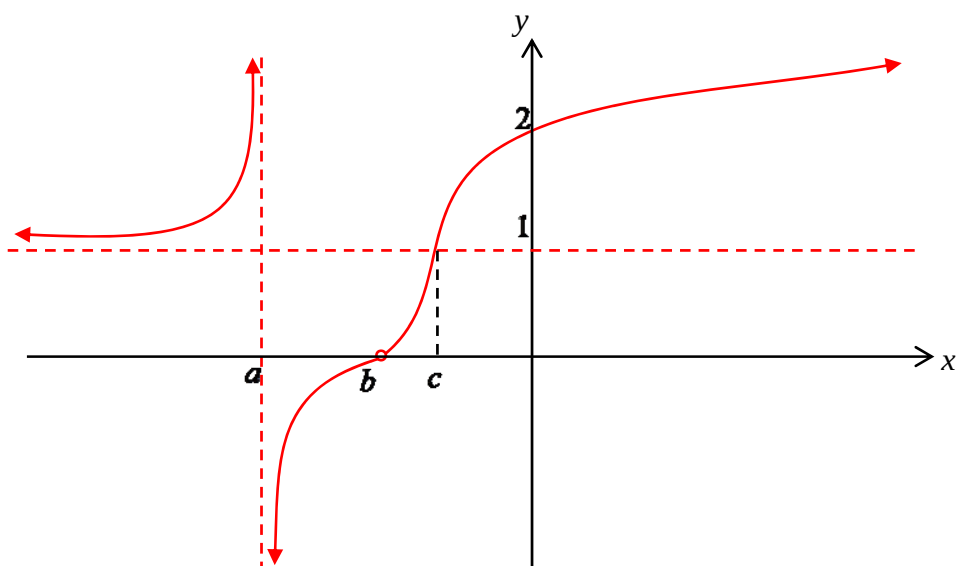
- (d) Given below is the graph of $y = f(x)$. The line $y = 1$ is a horizontal asymptote and $x = b$ is a vertical asymptote. The x -intercept is at $x = a$ and the y -intercept is at $y = \frac{1}{2}$.



Neatly sketch the graphs of the following showing all important information, including the location of a, b and c :

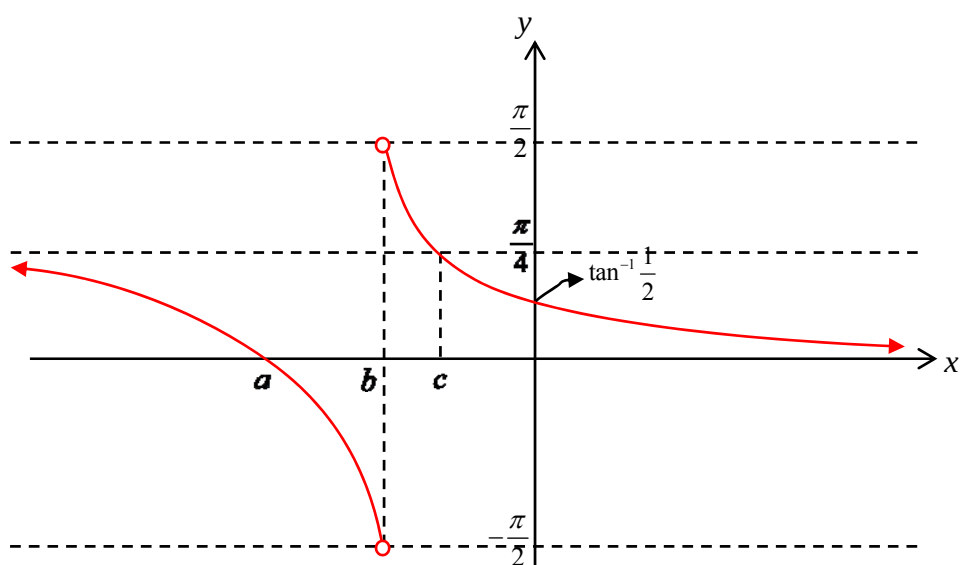
(i) $\frac{1}{f(x)}$

2



(ii) $\tan^{-1}(f(x))$

2



Question 13 (15 marks) Use a SEPARATE writing booklet.

(a) If $u = 6 + ki$ and $v = 4 + ki$, find k if $\arg(uv) = \frac{\pi}{4}$.

3

$$uv = (6 + ki)(4 + ki) = (24 - k^2) + i(10k)$$

$$\arg(uv) = \frac{\pi}{4}$$

$$\tan(\arg(uv)) = \frac{10k}{24 - k^2} = 1$$

$$10k = 24 - k^2$$

$$k^2 + 10k - 24 = 0$$

$$(k + 12)(k - 2) = 0$$

$$k = -12, 2$$

But $k \neq -12$ as this gives $uv = -120 - 120ki$ which is not in the first quadrant.

So $k = 2$ is the only solution.

(b) $P(a \sec \theta, b \tan \theta)$ is a point on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.

(i) Show that the tangent at P has the equation

2

$$\frac{x \sec \theta}{a} - \frac{y \tan \theta}{b} = 1$$

$$x = a \sec \theta \Rightarrow \frac{dx}{d\theta} = a \sec \theta \tan \theta$$

$$y = b \tan \theta \Rightarrow \frac{dy}{d\theta} = b \sec^2 \theta$$

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{b \sec^2 \theta}{a \sec \theta \tan \theta} = \frac{b \sec \theta}{a \tan \theta}$$

Equation of tangent at P is:

$$y - b \tan \theta = \frac{b \sec \theta}{a \tan \theta} (x - a \sec \theta)$$

$$ay \tan \theta - ab \tan^2 \theta = bx \sec \theta - ab \sec^2 \theta$$

$$bx \sec \theta - ay \tan \theta = ab(\sec^2 \theta - \tan^2 \theta) \quad \div ab$$

$$\frac{x \sec \theta}{a} - \frac{y \tan \theta}{b} = 1 \quad \text{as } \sec^2 \theta - \tan^2 \theta = 1$$

- (ii) The tangent at P meets the asymptotes of the hyperbola at Q and R .
Show that $PQ = PR$.

3

The asymptotes are $y = \pm \frac{b}{a}x$. Solving simultaneously with the tangent, we get

$$\frac{x \sec \theta}{a} - \frac{\left(\pm \frac{b}{a}x\right) \tan \theta}{b} = 1$$

$$\frac{x \sec \theta}{a} - \frac{\pm x \tan \theta}{a} = 1$$

$$\frac{x(\sec \theta \mp \tan \theta)}{a} = 1 \text{ or } x = \frac{a}{(\sec \theta \mp \tan \theta)}$$

$$\text{So, } Q = \left(\frac{a}{\sec \theta - \tan \theta}, \frac{b}{\sec \theta - \tan \theta} \right) \text{ using } y = \frac{b}{a}x \text{ and}$$

$$R = \left(\frac{a}{\sec \theta + \tan \theta}, \frac{-b}{\sec \theta + \tan \theta} \right) \text{ using } y = -\frac{b}{a}x$$

Now, if $PQ=PR$, then P is the midpoint of QR . Let M be the midpoint of QR .

$$x_M = \frac{1}{2} \left(\frac{a \sec \theta + \cancel{a \tan \theta} + a \sec \theta - \cancel{a \tan \theta}}{\sec^2 \theta - \tan^2 \theta} \right) \text{ but } \sec^2 \theta - \tan^2 \theta = 1$$

$$= \frac{2a \sec \theta}{2} = a \sec \theta$$

$$y_M = \frac{1}{2} \left(\frac{\cancel{b \sec \theta} + b \tan \theta - \cancel{b \sec \theta} + b \tan \theta}{\sec^2 \theta - \tan^2 \theta} \right) \text{ but } \sec^2 \theta - \tan^2 \theta = 1$$

$$= \frac{2b \tan \theta}{2} = b \tan \theta$$

$$\text{So, } M_{QR} = (a \sec \theta, b \tan \theta) = P \text{ or } PQ=PR$$

Alternate approaches use distance formula or geometric relationships using similar triangles or intercepts on parallel lines.

- (c) If α, β, γ are the roots of the equation $x^3 + 2x^2 + 3x - 4 = 0$, find a polynomial equation whose roots are:

(i) $\frac{1}{\alpha}, \frac{1}{\beta}$ and $\frac{1}{\gamma}$

2

An equation whose roots are $\frac{1}{\alpha}, \frac{1}{\beta}$ and $\frac{1}{\gamma}$ is given by:

$$\left(\frac{1}{x}\right)^3 + 2\left(\frac{1}{x}\right)^2 + 3\left(\frac{1}{x}\right) - 4 = 0 \text{ multiply by } x^3$$

$$1 + 2x + 3x^2 - 4x^3 = 0 \text{ or } 4x^3 - 3x^2 - 2x - 1 = 0$$

An equation with roots α^2, β^2 and γ^2 is given by:

$$(\sqrt{x})^3 + 2(\sqrt{x})^2 + 3(\sqrt{x}) - 4 = 0$$

$$x\sqrt{x} + 2x + 3\sqrt{x} - 4 = 0$$

$$(x+3)\sqrt{x} = 4-2x \quad \text{square both sides}$$

$$(x^2 + 6x + 9)x = 16 + 4x^2 - 16x$$

$$x^3 + 2x^2 + 25x - 16 = 0$$

(d) A sequence is defined by

3

$$a_1 = 5, a_2 = 13 \text{ and } a_{n+2} = 5a_{n+1} - 6a_n \text{ for all natural numbers } n.$$

Use Mathematical Induction to prove that $a_n = 2^n + 3^n$.

Let $S(n)$ be the statement $a_n = 2^n + 3^n$.

Test $n = 1$: $a_1 = 2^1 + 3^1 = 5$ So, $S(1)$ is true.

Test $n = 2$ $a_2 = 2^2 + 3^2 = 4 + 9 = 13$ So, $S(2)$ is true.

Assume $S(k)$ and $S(k+1)$ are true. i.e. assume $a_k = 2^k + 3^k$ and $a_{k+1} = 2^{k+1} + 3^{k+1}$

Prove $S(k+2)$ i.e. prove $a_{k+2} = 2^{k+2} + 3^{k+2}$.

$$\text{LHS} = a_{k+2}$$

$$= 5a_{k+1} - 6a_k \quad \text{using the given recursive formula}$$

$$= 5(2^{k+1} + 3^{k+1}) - 6(2^k + 3^k) \quad \text{by assumption}$$

$$= 5 \cdot 2^{k+1} + 5 \cdot 3^{k+1} - 6 \cdot 2^k - 6 \cdot 3^k$$

$$= 5 \cdot 2^{k+1} - 3 \cdot 2 \cdot 2^k + 5 \cdot 3^{k+1} - 2 \cdot 3 \cdot 3^k$$

$$= 5 \cdot 2^{k+1} - 3 \cdot 2^{k+1} + 5 \cdot 3^{k+1} - 2 \cdot 3^{k+1}$$

$$= 2 \cdot 2^{k+1} + 3 \cdot 3^{k+1}$$

$$= 2^{k+2} + 3^{k+2}$$

$$= \text{RHS}$$

Hence, $S(k+2)$ is true.

So, $S(n)$ is true by Mathematical Induction.

Question 14 (15 marks) Use a SEPARATE writing booklet

(a) Using the substitution $t = \tan \frac{x}{2}$, or otherwise, evaluate

3

$$\int_0^{\frac{\pi}{3}} \frac{1}{1 + \cos x - \sin x} dx$$

Let $t = \tan \frac{x}{2}$ then $x = 2 \tan^{-1} t$ and $dx = \frac{2dt}{1+t^2}$

Also, $\cos x = \frac{1-t^2}{1+t^2}$ and $\sin x = \frac{2t}{1+t^2}$

x	0	$\frac{\pi}{3}$
t	0	$\frac{1}{\sqrt{3}}$

$$\begin{aligned} \int_0^{\frac{\pi}{3}} \frac{1}{1 + \cos x - \sin x} dx &= \int_0^{\frac{1}{\sqrt{3}}} \frac{1}{1 + \frac{1-t^2}{1+t^2} - \frac{2t}{1+t^2}} \times \frac{2dt}{1+t^2} \\ &= \int_0^{\frac{1}{\sqrt{3}}} \frac{2dt}{1+t^2 + 1-t^2 - 2t} \\ &= \int_0^{\frac{1}{\sqrt{3}}} \frac{2dt}{2-2t} = \int_0^{\frac{1}{\sqrt{3}}} \frac{dt}{1-t} \\ &= \left[-\log(1-t) \right]_0^{\frac{1}{\sqrt{3}}} \\ &= -\log\left(1 - \frac{1}{\sqrt{3}}\right) + \log 1 \\ &= -\log\left(\frac{\sqrt{3}-1}{\sqrt{3}}\right) = \log\left(\frac{\sqrt{3}}{\sqrt{3}-1}\right) \end{aligned}$$

(b) (i) Use integration by parts to find $\int x \tan^{-1} x \, dx$.

2

$\int x \tan^{-1} x \, dx = \frac{x^2}{2} \tan^{-1} x - \int \frac{x^2}{2} \cdot \frac{1}{1+x^2} dx$ using integration by parts

$$= \frac{1}{2} \left[x^2 \tan^{-1} x - \int \frac{x^2 + 1 - 1}{1+x^2} dx \right]$$

$$= \frac{1}{2} \left[x^2 \tan^{-1} x - \int \left(1 - \frac{1}{1+x^2} \right) dx \right]$$

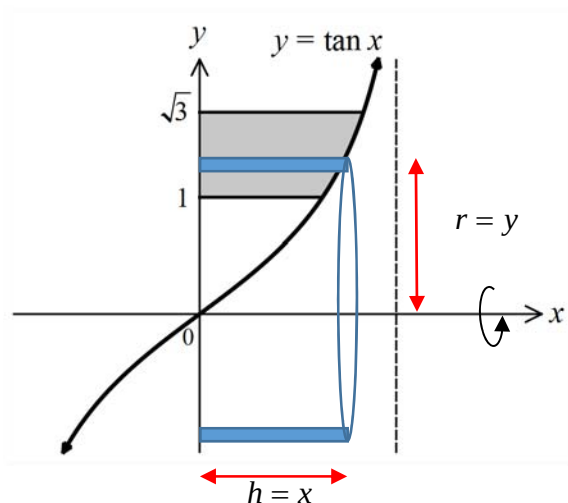
$$= \frac{1}{2} \left[x^2 \tan^{-1} x - x + \tan^{-1} x \right] + C$$

$$= \frac{1}{2} \left[(x^2 + 1) \tan^{-1} x - x \right] + C$$

$$\begin{aligned} u &= \tan^{-1} x & v' &= x \\ u' &= \frac{1}{1+x^2} & v &= \frac{x^2}{2} \end{aligned}$$

- (ii) The region bounded by the curve $y = \tan x$, the lines $y = 1$, $y = \sqrt{3}$ and the y -axis is rotated about the x -axis.

2

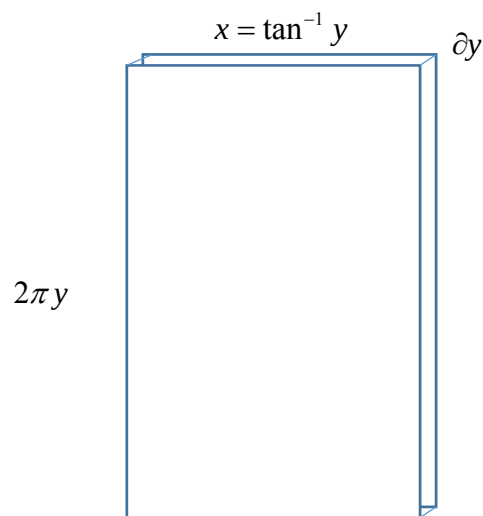


Use the method of cylindrical shells to find the volume of the solid of revolution formed.

Radius of shell is y and height of cylindrical shell is $x = \tan^{-1} y$. Thickness of shell is δy .

Volume of cylindrical shell is $\delta V = 2\pi y \tan^{-1} y \delta y$.

$$\begin{aligned}
 V &= \lim_{\delta y \rightarrow 0} \sum_{y=1}^{\sqrt{3}} \delta V \\
 &= 2\pi \int_1^{\sqrt{3}} y \tan^{-1} y \, dy \\
 &= 2\pi \frac{1}{2} \left[(y^2 + 1) \tan^{-1} y - y \right]_1^{\sqrt{3}} \\
 &= \pi \left[(4 \tan^{-1} \sqrt{3} - \sqrt{3}) - (2 \tan^{-1} 1 - 1) \right] \\
 &= \pi \left(\frac{4\pi}{3} - \sqrt{3} - 2 \frac{\pi}{4} + 1 \right) \\
 &= \pi \left(\frac{5\pi}{6} - \sqrt{3} + 1 \right) \text{ units}^3
 \end{aligned}$$



Cylindrical shell as rectangular prism

- (c) (i) Show that $a^2 + 9b^2 \geq 6ab$, where a and b are real numbers.

1

$$a^2 + 9b^2 - 6ab = (a - 3b)^2 \geq 0 \quad \text{for all real } a, b \text{ (perfect square)}$$

$$\therefore a^2 + 9b^2 \geq 6ab$$

- (ii) Hence, or otherwise, show that $a^2 + 5b^2 + 9c^2 \geq 3(ab + bc + ac)$.

2

$$a^2 + 9b^2 \geq 6ab \text{ from (i)}$$

$$\text{Similarly, } b^2 + 9c^2 \geq 6bc \text{ and } a^2 + 9c^2 \geq 6ac$$

Adding the three results we have:

$$2a^2 + 10b^2 + 18c^2 \geq 6(ab + bc + ac)$$

$$a^2 + 5b^2 + 9c^2 \geq 3(ab + bc + ac)$$

(iii) Hence if $a > b > c > 0$, show that $a^2 + 5b^2 + 9c^2 > 9bc$.

1

$$a^2 + 5b^2 + 9c^2 \geq 3(ab + ac + bc) \quad \text{from (ii)}$$

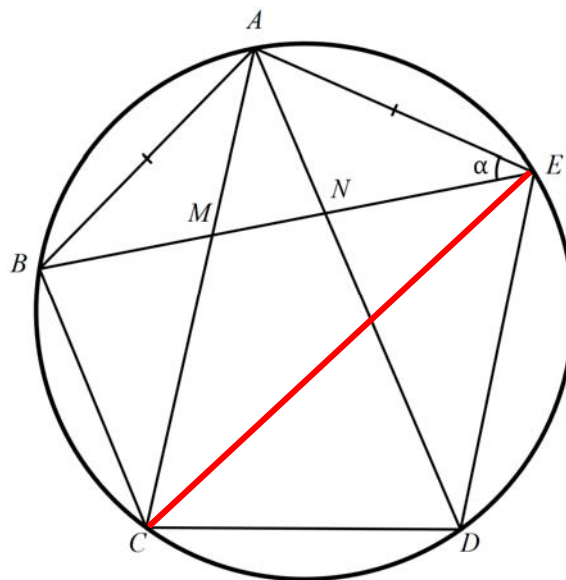
$$> 3(bc + bc + bc) \quad \text{as } a > c \Rightarrow ab > bc \text{ and } a > b \Rightarrow ac > bc$$

$$= 3(3bc)$$

$$= 9bc$$

$$a^2 + 5b^2 + 9c^2 > 9bc$$

(d) $ABCDE$ is a pentagon inscribed in a circle and $AB = AE$.
 BE meets AC and AD at M and N respectively. Let $\angle BEA = \alpha$.



(i) Explain why $\angle ACE = \alpha$.

1

Join CE .

AB and AE are equal chords and hence subtend equal angles at the centre and hence at the circumference. Therefore, $\angle AEB = \angle ACE = \alpha$

(ii) Prove that $CDNM$ is a cyclic quadrilateral.

3

Let $\angle ECD = \beta$. Then $\angle ECD = \angle EAD = \beta$ (angles in same segment on chord ED)

$\angle MCD = \alpha + \beta$ (adding adjacent angles)

$\angle ANM = \alpha + \beta$ (exterior angle of $\triangle ANE$)

$\therefore \angle ANM = \angle MCD$

Then, $CDNM$ is cyclic (exterior angle is equal to opposite interior angle)

Question 15 (15 marks) Use a SEPARATE writing booklet.

- (a) Given that $p(x) = x^3 - 3ax^2 + b$ has a double zero, where a and b are non-zero real numbers, show that $4a^3 - b = 0$. **2**

Let $x = \alpha$ be the double zero. Then $p(\alpha) = p'(\alpha) = 0$ by the Multiple Root Theorem

$$p'(x) = 3x^2 - 6ax$$

$$p'(\alpha) = 3\alpha^2 - 6a\alpha = 0$$

$$3\alpha(\alpha - 2a) = 0$$

$$\alpha = 2a \quad \text{Note } \alpha \neq 0 \text{ as } P(0) = b \neq 0$$

$$p(\alpha) = \alpha^3 - 3a\alpha^2 + b = 0$$

$$(2a)^3 - 3a(2a)^2 + b = 0$$

$$8a^3 - 12a^3 + b = 0$$

$$-4a^3 + b = 0 \quad \text{or} \quad 4a^3 - b = 0$$

- (b) $P\left(cp, \frac{c}{p}\right)$ lies on the hyperbola $xy = c^2$.

- (i) Show that the equation of the normal to the hyperbola at P is given by **2**

$$px - \frac{y}{p} = c\left(p^2 - \frac{1}{p^2}\right).$$

$$x = cp \Rightarrow \frac{dx}{dp} = c$$

$$y = \frac{c}{p} \Rightarrow \frac{dy}{dp} = -\frac{c}{p^2}$$

$$\frac{dy}{dx} = \frac{\frac{dy}{dp}}{\frac{dx}{dp}} = \frac{-c/p^2}{c} = -\frac{1}{p^2}$$

Gradient of normal is p^2 .

Equation of normal is:

$$y - \frac{c}{p} = p^2(x - cp) \quad \text{divide by } p$$

$$\frac{y}{p} - \frac{c}{p^2} = px - cp^2$$

$$px - \frac{y}{p} = cp^2 - \frac{c}{p^2}$$

$$px - \frac{y}{p} = c\left(p^2 - \frac{1}{p^2}\right)$$

(ii) The equation of the tangent at P is $x + p^2y = 2cp$. DO NOT PROVE THIS.

3

The normal at P cuts the x -axis at Q and the tangent at P cuts the y -axis at R .
 M is the midpoint of QR .

Find the equation of the locus of M as P moves on the hyperbola.

$$Q = \left(c \left(p - \frac{1}{p^3} \right), 0 \right) \text{ and } R = \left(0, \frac{2c}{p} \right)$$

$$M = \left(\frac{c}{2} \left(p - \frac{1}{p^3} \right), \frac{c}{p} \right)$$

For locus of M ,

$$x = \frac{c}{2} \left(p - \frac{1}{p^3} \right) \quad (1) \quad \text{and} \quad y = \frac{c}{p} \quad (2)$$

From (2), $p = \frac{c}{y}$. Sub into (1)

$$x = \frac{c}{2} \left(\frac{c}{y} - \frac{y^3}{c^3} \right)$$

$$x = \frac{c^2}{2y} - \frac{y^3}{2c^2} \quad \text{multiply by } 2c^2y$$

$$2c^2xy = c^4 - y^4$$

(c) Let
$$I = \int_{\frac{1}{a}}^a \frac{f(x)}{x \left(f(x) + f\left(\frac{1}{x}\right) \right)} dx.$$

(i) Use a suitable substitution to show that
$$I = \int_{\frac{1}{a}}^a \frac{f\left(\frac{1}{x}\right)}{x \left(f(x) + f\left(\frac{1}{x}\right) \right)} dx.$$

2

Let $u = \frac{1}{x}$. Then $du = -\frac{1}{x^2} dx$

When $x = \frac{1}{a}$, $u = a$ and when $x = a$, $u = \frac{1}{a}$

$$f(x) = f\left(\frac{1}{u}\right) \text{ and } f\left(\frac{1}{x}\right) = f(u)$$

$$\begin{aligned}
\int_{\frac{1}{a}}^a \frac{f(x)}{x \left(f(x) + f\left(\frac{1}{x}\right) \right)} dx &= \int_a^{\frac{1}{a}} \frac{f\left(\frac{1}{u}\right)}{\frac{1}{u} \left(f\left(\frac{1}{u}\right) + f(u) \right)} \left(-\frac{1}{u^2} du \right) \\
&= \int_{\frac{1}{a}}^a \frac{f\left(\frac{1}{u}\right)}{u \left(f\left(\frac{1}{u}\right) + f(u) \right)} du \\
&= \int_{\frac{1}{a}}^a \frac{f\left(\frac{1}{x}\right)}{x \left(f\left(\frac{1}{x}\right) + f(x) \right)} dx \quad \text{as } u \text{ is a dummy variable}
\end{aligned}$$

(ii) Hence, or otherwise, evaluate $\int_{\frac{1}{2}}^2 \frac{\sin x}{x \left(\sin x + \sin \frac{1}{x} \right)} dx$. 2

Let $I = \int_{\frac{1}{2}}^2 \frac{\sin x}{x \left(\sin x + \sin \frac{1}{x} \right)} dx = \int_{\frac{1}{2}}^2 \frac{\sin \frac{1}{x}}{x \left(\sin x + \sin \frac{1}{x} \right)} dx$ using result in (i)

$$2I = \int_{\frac{1}{2}}^2 \frac{\sin x}{x \left(\sin x + \sin \frac{1}{x} \right)} dx + \int_{\frac{1}{2}}^2 \frac{\sin \frac{1}{x}}{x \left(\sin x + \sin \frac{1}{x} \right)} dx$$

$$= \int_{\frac{1}{2}}^2 \frac{\cancel{\sin x} + \cancel{\sin \frac{1}{x}}}{x \left(\cancel{\sin x} + \cancel{\sin \frac{1}{x}} \right)} dx$$

$$= \int_{\frac{1}{2}}^2 \frac{1}{x} dx$$

$$= [\log_e x]_{\frac{1}{2}}^2$$

$$= \log 2 - \log \frac{1}{2} = \log 2 + \log 2$$

$$= 2 \log 2$$

$$I = \log 2$$

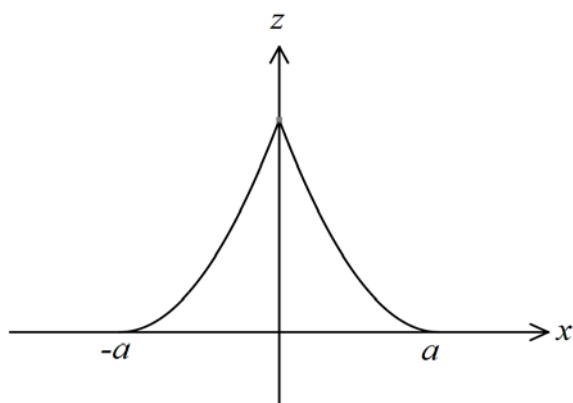
- (d) (i) Find the area bounded by the curve $y = (x - a)^2$ and the coordinate axes, where $a > 0$.

1

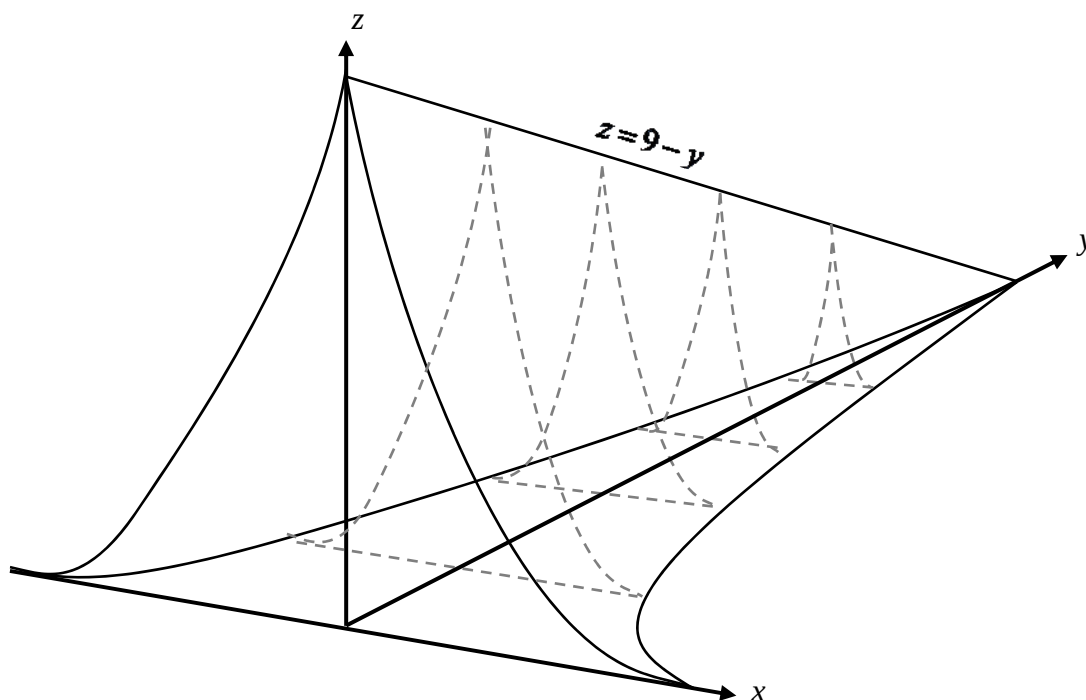
$$\begin{aligned} A &= \int_0^a (x - a)^2 dx \\ &= \left[\frac{(x - a)^3}{3} \right]_0^a \\ &= 0 - \left(\frac{-a^3}{3} \right) = \frac{a^3}{3} \end{aligned}$$

- (ii) Cross-sections of a solid perpendicular to the base are sections of two parabolas $z = (x - a)^2$ and $z = (x + a)^2$ as shown below.

3

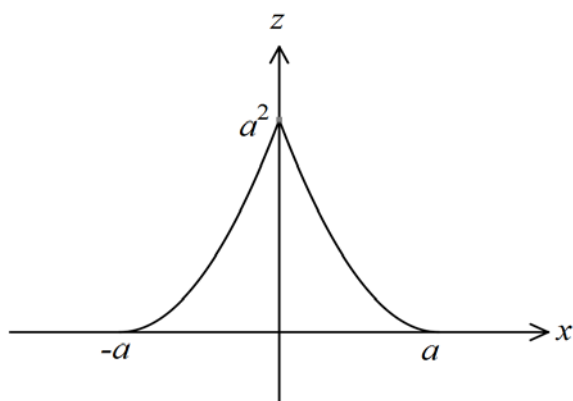


The heights of the cross-sections are bounded by the line $z = 9 - y$.



Find the volume of the solid.

Typical element:



Area of typical element is $\frac{2a^3}{3}$ from (i), where a is the x -intercept of the parabolic segment.

The height of the typical element is $9 - y$ which equates to a^2 (the y -intercept of the parabolic segment). So $a = (9 - y)^{\frac{1}{2}}$.

So, area of the typical element is $\frac{2}{3}(9 - y)^{\frac{3}{2}}$.

Volume of typical element is $\delta V = \frac{2}{3}(9 - y)^{\frac{3}{2}} \delta y$

$$\begin{aligned}
 V &= \lim_{\delta y \rightarrow 0} \sum_{y=0}^9 \delta V \\
 &= \int_0^9 \frac{2}{3}(9 - y)^{\frac{3}{2}} dy \\
 &= \frac{2}{3} \left[-\frac{2}{5}(9 - y)^{\frac{5}{2}} \right]_0^9 \\
 &= \frac{-4}{15}(0 - 3^5) \\
 &= \frac{324}{5} \text{ units}^3
 \end{aligned}$$

Question 16 (15 marks) Use a SEPARATE writing booklet.

(a) Consider $I_n = \int (x^2 + 1)^{-n} dx$, $n > 0$.

(i) Show that $I_{n+1} = \frac{x(x^2 + 1)^{-n}}{2n} + \frac{2n-1}{2n} I_n$.

3

Using integration by parts,

$$\begin{aligned} I_n &= \int 1 \cdot (x^2 + 1)^{-n} dx \\ &= x(x^2 + 1)^{-n} - \int -2nx^2 (x^2 + 1)^{-(n+1)} dx \end{aligned} \quad \left| \quad \begin{aligned} u &= (x^2 + 1)^{-n} & v' &= 1 \\ u' &= -2nx(x^2 + 1)^{-(n+1)} & v &= x \end{aligned} \right.$$

$$\begin{aligned} I_n &= x(x^2 + 1)^{-n} + 2n \int (x^2 + 1 - 1)(x^2 + 1)^{-(n+1)} dx \\ &= x(x^2 + 1)^{-n} + 2n \int \left[(x^2 + 1)^{-n} - (x^2 + 1)^{-(n+1)} \right] dx \\ &= x(x^2 + 1)^{-n} + 2nI_n - 2nI_{n+1} \end{aligned}$$

Rearranging, we have

$$2nI_{n+1} = x(x^2 + 1)^{-n} + (2n-1)I_n$$

$$I_{n+1} = \frac{x(x^2 + 1)^{-n}}{2n} + \frac{(2n-1)}{2n} I_n$$

(ii) Find I_2 .

1

$$\begin{aligned} I_2 &= \frac{x(x^2 + 1)^{-1}}{2} + \frac{1}{2} I_1 \quad \text{using (i)} \\ &= \frac{x}{2(x^2 + 1)} + \frac{1}{2} \int \frac{dx}{x^2 + 1} \\ &= \frac{x}{2(x^2 + 1)} + \frac{1}{2} \tan^{-1} x + C \end{aligned}$$

(b) (i) Using the result $\cos(A+B) - \cos(A-B) = -2\sin A \sin B$, or otherwise,

2

show that $\sin \theta \sum_{r=1}^n \sin 2r\theta = \sin(n+1)\theta \sin n\theta$.

$$\begin{aligned}
 \sin \theta \sum_{r=1}^n \sin 2r\theta &= \sum_{r=1}^n \sin \theta \sin 2r\theta = -\frac{1}{2} \sum_{r=1}^n -2\sin \theta \sin 2r\theta \\
 &= -\frac{1}{2} \sum_{r=1}^n [\cos(2r+1)\theta - \cos(2r-1)\theta] \quad \text{using the given result} \\
 &= -\frac{1}{2} \left[\sum_{r=1}^n \cos(2r+1)\theta - \sum_{r=1}^n \cos(2r-1)\theta \right] \\
 &= -\frac{1}{2} \left[\sum_{r=1}^n \cos(2r+1)\theta - \sum_{r=0}^{n-1} \cos(2r+1)\theta \right] \\
 &= -\frac{1}{2} \left[\sum_{r=1}^{n-1} \cos(2r+1)\theta + \cos(2n+1)\theta - \cos \theta - \sum_{r=1}^{n-1} \cos(2r+1)\theta \right] \\
 &= -\frac{1}{2} [\cos(2n+1)\theta - \cos \theta] \\
 &= -\frac{1}{2} [\cos(n\theta + \theta + n\theta) - \cos(n\theta + \theta - n\theta)] \\
 &= -\frac{1}{2} [-2\sin(n\theta + \theta)\sin n\theta] \quad \text{using the given result} \\
 &= \sin(n+1)\theta \sin n\theta \quad \text{as required}
 \end{aligned}$$

(ii) Hence show that $\sum_{r=1}^8 \sin \frac{r\pi}{9} = \cot \frac{\pi}{18}$.

2

Using $\theta = \frac{\pi}{18}$ and $n = 8$ in (i) we have

$$\begin{aligned}
 \sin \frac{\pi}{18} \sum_{r=1}^8 \sin \frac{2r\pi}{18} &= \sin 9 \left(\frac{\pi}{18} \right) \sin 8 \left(\frac{\pi}{18} \right) \\
 \sin \frac{\pi}{18} \sum_{r=1}^8 \sin \frac{r\pi}{9} &= \sin \frac{\pi}{2} \sin \left(\frac{4\pi}{9} \right) \\
 \sum_{r=1}^8 \sin \frac{r\pi}{9} &= \frac{\sin \frac{\pi}{2} \sin \left(\frac{4\pi}{9} \right)}{\sin \frac{\pi}{18}} = \frac{\sin \left(\frac{4\pi}{9} \right)}{\sin \frac{\pi}{18}} \\
 &= \frac{\cos \left(\frac{\pi}{2} - \frac{4\pi}{9} \right)}{\sin \frac{\pi}{18}} = \frac{\cos \frac{\pi}{18}}{\sin \frac{\pi}{18}} \\
 &= \cot \frac{\pi}{18}
 \end{aligned}$$

(c) (i) Show that $\log_e x \leq x - 1$ for $x > 0$.

2

Let $f(x) = \log x - x + 1$.

$$f'(x) = \frac{1}{x} - 1 = \frac{1-x}{x}$$

Stationary point at $f'(x) = 0 \Rightarrow x = 1$

$f(1) = 0$. So, stationary point at $(1, 0)$.

$$f''(x) = -\frac{1}{x^2} < 0 \text{ for all } x.$$

So the function is always concave down and $(1, 0)$ is a maximum turning point.

$\therefore f(x) = \log x - x + 1 \leq 0$ for all x in its domain ie for $x > 0$.

So $\log x - x + 1 \leq 0$ or $\log x \leq x - 1$

(ii) Show that $\log_e \left(\frac{c_1 c_2 \dots c_n}{\mu^n} \right) \leq \frac{c_1 + c_2 + \dots + c_n}{\mu} - n$, where $c_1, c_2, \dots, c_n > 0$ and $\mu > 0$. 2

$$\begin{aligned} \log \left(\frac{c_1 c_2 \dots c_n}{\mu^n} \right) &= \log \left(\frac{c_1}{\mu} \cdot \frac{c_2}{\mu} \dots \frac{c_n}{\mu} \right) \\ &= \log \left(\frac{c_1}{\mu} \right) + \log \left(\frac{c_2}{\mu} \right) + \dots + \log \left(\frac{c_n}{\mu} \right) \\ &\leq \left(\frac{c_1}{\mu} - 1 \right) + \left(\frac{c_2}{\mu} - 1 \right) + \dots + \left(\frac{c_n}{\mu} - 1 \right) \text{ using (i)} \\ &= \frac{c_1}{\mu} + \frac{c_2}{\mu} + \dots + \frac{c_n}{\mu} - n \\ &= \frac{c_1 + c_2 + \dots + c_n}{\mu} - n \end{aligned}$$

(iii) Hence if $\mu = \frac{c_1 + c_2 + \dots + c_n}{n}$, show that $\sqrt[n]{c_1 c_2 \dots c_n} \leq \frac{c_1 + c_2 + \dots + c_n}{n}$. 2

Using $\mu = \frac{c_1 + c_2 + \dots + c_n}{n}$ in the result from (ii)

$$\begin{aligned} \log \left(\frac{c_1 c_2 \dots c_n}{\left(\frac{c_1 + c_2 + \dots + c_n}{n} \right)^n} \right) &\leq \frac{c_1 + c_2 + \dots + c_n}{\left(\frac{c_1 + c_2 + \dots + c_n}{n} \right)} - n \\ \log \left(\frac{n^n c_1 c_2 \dots c_n}{(c_1 + c_2 + \dots + c_n)^n} \right) &\leq n - n = 0 \end{aligned}$$

$$\frac{n^n c_1 c_2 \dots c_n}{(c_1 + c_2 + \dots + c_n)^n} \leq 1$$

$$n^n c_1 c_2 \dots c_n \leq (c_1 + c_2 + \dots + c_n)^n$$

$$c_1 c_2 \dots c_n \leq \left(\frac{c_1 + c_2 + \dots + c_n}{n} \right)^n$$

$$\sqrt[n]{c_1 c_2 \dots c_n} \leq \frac{c_1 + c_2 + \dots + c_n}{n}$$

$$\frac{c_1 + c_2 + \dots + c_n}{n} \geq \sqrt[n]{c_1 c_2 \dots c_n}$$

(iv) Hence use part (iii) to find a lower bound for $\frac{101}{103} + \frac{103}{105} + \frac{105}{107} + \dots + \frac{197}{199} + \frac{199}{101}$. **1**

Using result (iii) with $c_1 = \frac{101}{103}; c_2 = \frac{103}{105}; c_3 = \frac{105}{107} \dots c_{50} = \frac{199}{101}$

$$\frac{\frac{101}{103} + \frac{103}{105} + \dots + \frac{197}{199} + \frac{199}{101}}{50} \geq \sqrt[50]{\frac{101}{103} \times \frac{103}{105} \times \dots \times \frac{197}{199} \times \frac{199}{101}} = 1$$

$$\frac{101}{103} + \frac{103}{105} + \dots + \frac{197}{199} + \frac{199}{101} \geq 50$$

End of solutions